Author Contributions Checklist Form

This form documents the artifacts associated with the article (i.e., the data and code supporting the computational findings) and describes how to reproduce the findings.

Part 1: Data

☐ This paper **does not** involve analysis of external data (i.e., no data are used or the only data are generated by the authors via simulation in their code).

☒ I certify that the author(s) of the manuscript have legitimate access to and permission to use the data used in this manuscript.

Abstract

The empirical analyses use two datasets. The first is the Beijing PM2.5 dataset, which contains hourly environmental observations from March 5, 2013, to December 31, 2015. The second is the chemical sensor dataset, which contains readings from 16 chemical sensors exposed to mixtures of ethylene and methane in air at varying concentrations.

Availability

☒ Data **are** publicly available
☐ Data **cannot be made** publicly available

If the data are publicly available, see the *Publicly available data* section. Otherwise, see the *Non-publicly available data* section, below.

Publicly available data

☒ Data are available online at: https://weatherspark.com; https://archive.ics.uci.edu/dataset/322
☐ Data are available as part of the paper's supplementary material.
☐ Data are publicly available by request, following the process described here:

☐ Data are available through some other mechanism, described here:

Non-publicly available data

Discussion of lack of publicly available data:

Description

File format(s)

☒ CSV or other plain text: pm2.5.csv; ethylene_methane.txt
☐ Software-specific binary format (.Rda, Python pickle, etc.):
☐ Standardized binary format (e.g., netCDF, HDF5, etc.):
☐ Other (described here):

Click or tap here to enter text.

Data dictionary

☐ Provided by the authors in the following file(s):
☐ Data file(s) is (are) self-describiing (e.g., netCDF, HDF5, Zarr files)
☒ Available at the following URL:

https://weatherspark.com; https://archive.ics.uci.edu/dataset/322

Additional information (optional)

Click or tap here to enter text.

Part 2: Code

Abstract

We provide R code for all simulation studies and empirical analyses in the paper. Each program performs 1,000 replications and produces results under the same settings as those reported in the paper.

Description

Code format(s)

☒ Script files

☒ R ☐ Python ☐ Julia ☐ MATLAB

☐ Other:

☐ Package

☐ R ☐ Python ☐ Julia ☐ MATLAB toolbox

☐ Other:

☐ Reproducible report

☐ Quarto or R Markdown ☐ Jupyter notebook

☐ Other:

☐ Shell script

☐ Other (described here):

Supporting software requirements

Version of primary software used

R version 4.1.0

Libraries and dependencies used by the code

Libraries: TeachingSampling; quadprog (4.1.1); glmnet; (4.1.3); ncvreg (4.1.3)

Supporting system/hardware requirements (optional)

Parallelization used

☒ No parallel code used
☐ Multi-core parallelization on a single machine/node
Number of cores used:
☐ Multi-machine/multi-node parallelization
Number of nodes and cores used:

License

☒ MIT License (default)
☐ BSD
☐ GPL v3.0
☐ Creative Commons
☐ Other (described here):

Additional information (optional)

Part 3: Reproducibility workflow

Scope

The provided workflow reproduces:

☐ Any numbers provided in text in the paper

☒ The computational method(s) presented in the paper (i.e., code is provided that implements the method(s))

☒ All tables and figures in the paper

☐ Selected tables and figures in the paper, as explained and justified here:

Click or tap here to enter text.

Workflow details

Location

The workflow is available:

☐ As part of the paper's supplementary material

☒ In this Git repository:

☐ Other:

Format(s)

☒ Single master code file

☐ Wrapper (shell) script(s)

☐ Self-contained Quarto, R Markdown file, Jupyter notebook, or other literate programming approach

☐ Text file (e.g., a readme-style file) that documents workflow

☐ Makefile

☒ Other (more detail in 'Instructions' below)

Instructions

1. Five R scripts are provided. For the simulation studies, run scripts Sim1–Sim3 directly. For the empirical analyses, download the datasets, place them in the R working directory, and run scripts real1–real2.

2. Before running the scripts, install the required R packages: TeachingSampling, quadprog, glmnet, and ncvreg.
3. Each script runs independently. Adjust its settings as specified in the paper to generate the corresponding tables and figures.

Expected run-time

Approximate time needed to reproduce the analyses on a standard desktop machine:

☐ <1 minute

☐ 1-10 minutes

☐ 10-60 minutes

☒ 1-8 hours

☐ >8 hours

☐ Not feasible to run on a desktop machine, as described here:

Additional documentation (optional)

Notes (optional)